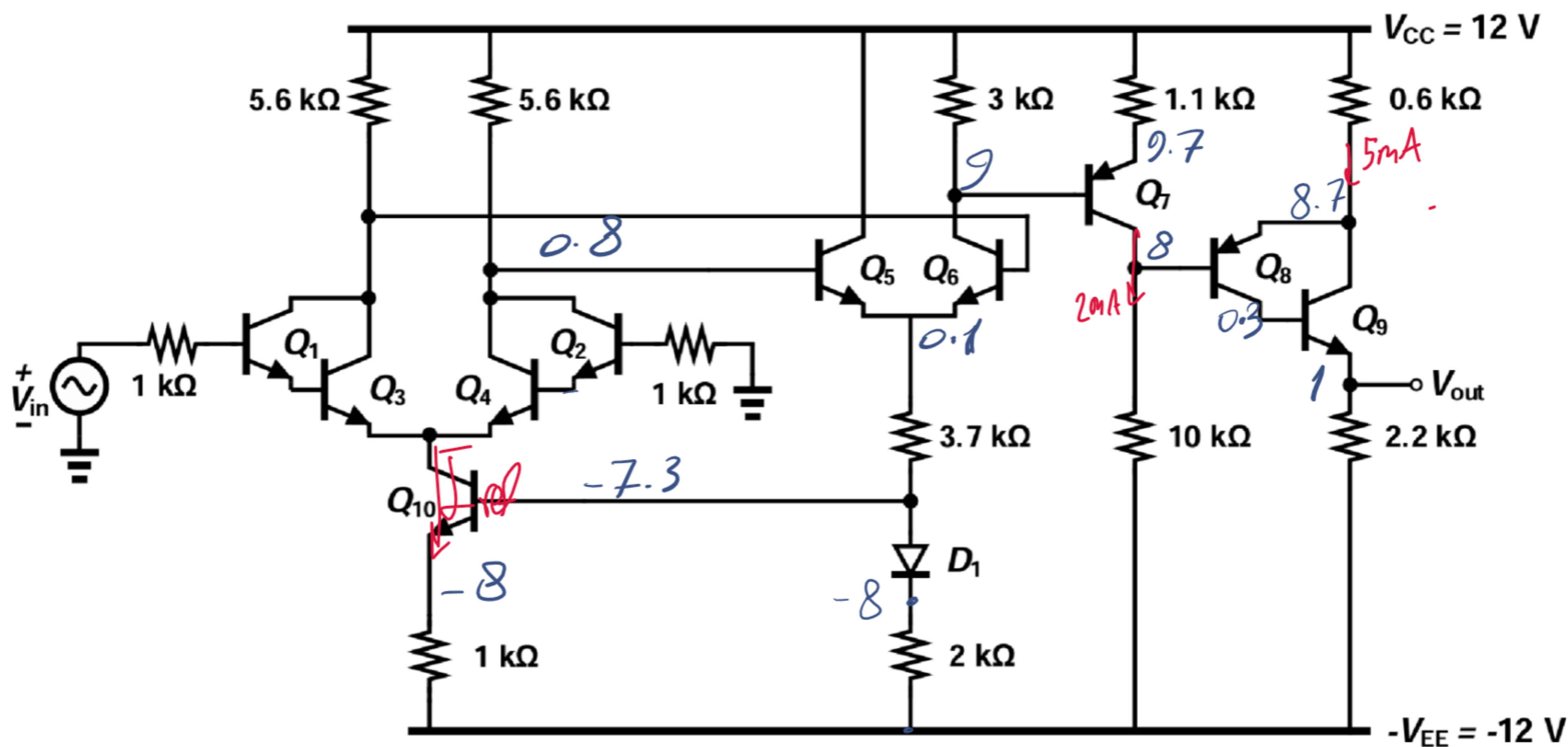




$$\bar{I}_{C4} = \frac{\bar{I}_{ref}}{2} = \beta \bar{I}_{C2} \rightarrow \bar{I}_{C2} = \frac{\bar{I}_{ref}}{2\beta}$$

$$V_{B5} = 12 - R_{C2} \frac{\bar{I}_{ref}}{2}$$

$$10.6 - (2.8) \bar{I}_{ref} - 2 \bar{I}_{C5} (5.7) = -12$$

$$22.6 = 2.8 \bar{I}_{ref} + 2 \bar{I}_{C5} (5.7)$$

$$-12 + 4 \bar{I}_{C5} + 0.7 = V_{Bb} = 0.7 + \bar{I}_{ref} - 12$$

$$\bar{I}_{ref} = 4 \bar{I}_{C5}$$

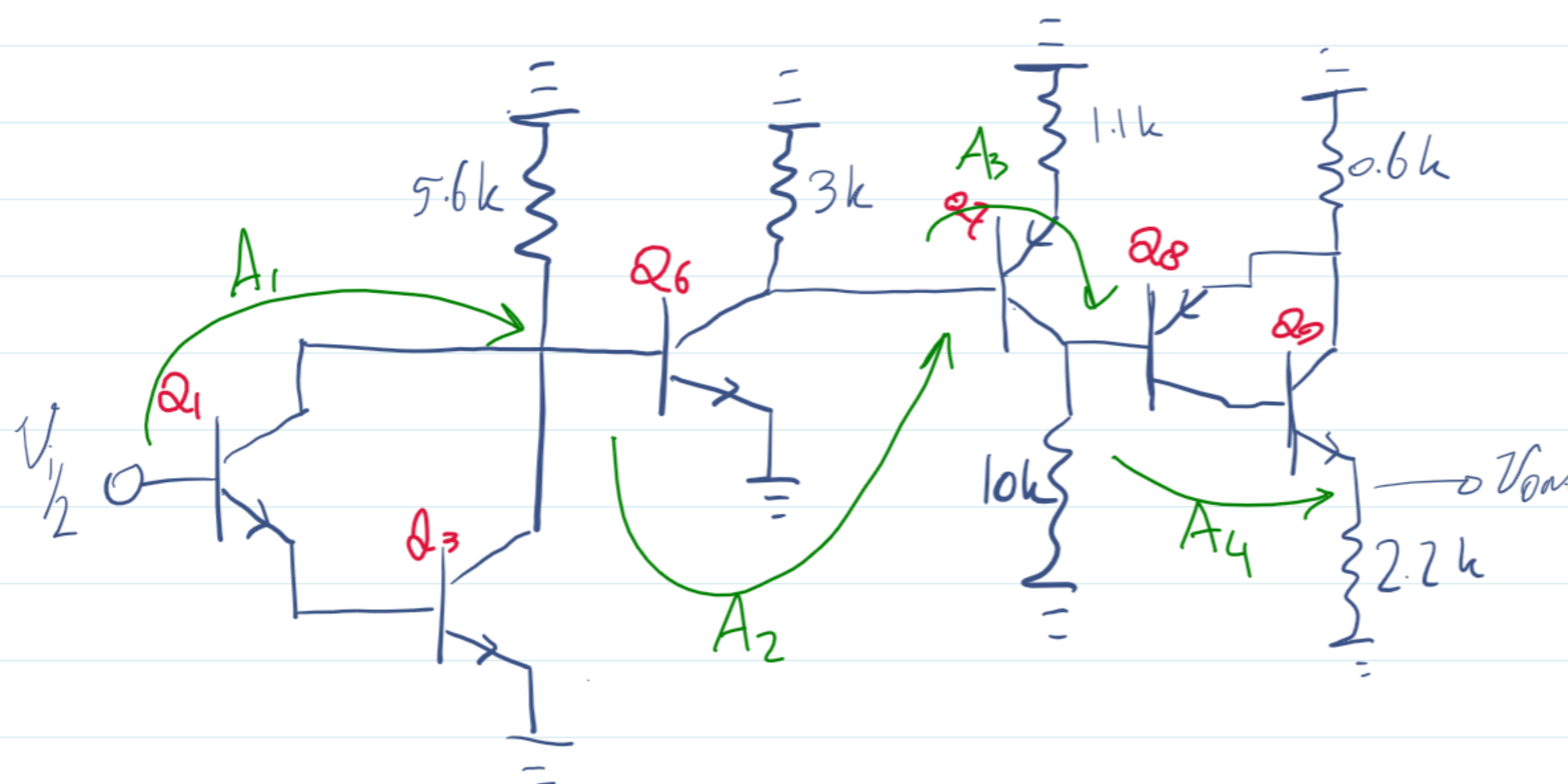
$$\bar{I}_{ref} = 4 \text{ mA}$$

$$\bar{I}_{C5} = 1 \text{ mA}$$

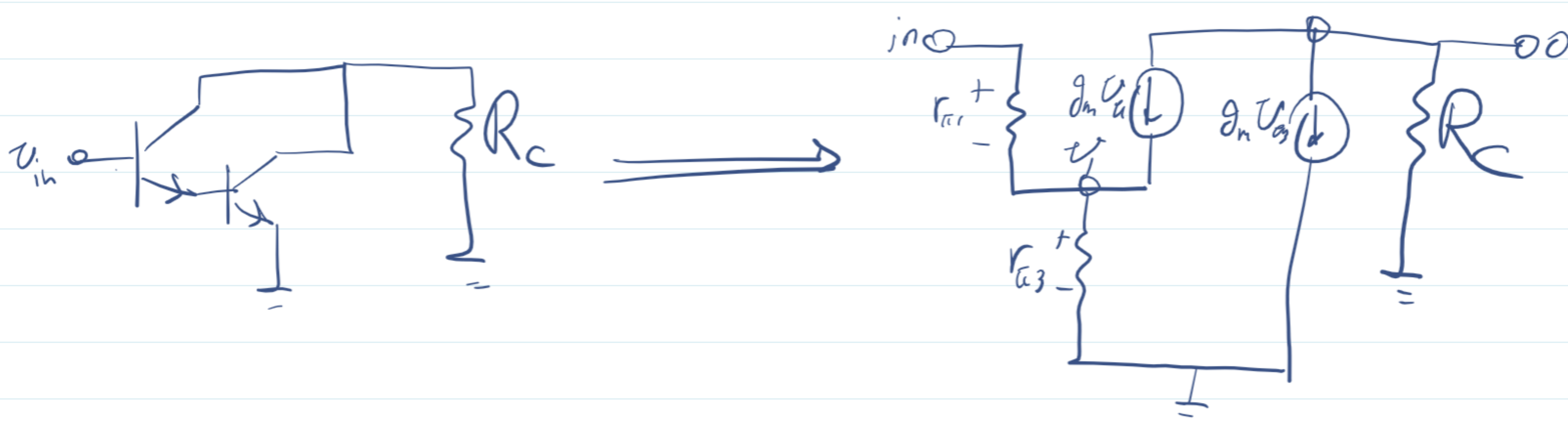
	Q ₁	Q ₃	Q ₆	Q ₇	Q ₈	Q ₉
$\bar{I}_{C(mA)}$	0.01	2	1	2	0.025	5
$g_m(mS^{-1})$	0.4	80	40	80	1	200
r_π	500	2.5	5	1.25	100	1
r_o	-	-	-	-	-	-

$$r_{o,Q10} = 25 \text{ k}\Omega$$

تفاضلی:



$$R_{in9} = r_{\pi9} + \beta_n (2.2k) = 440k\Omega$$



$$V_i \left(\frac{1}{r_{\pi3}} + \frac{1}{r_{\pi1}} + g_{m1} \right) - V_{in} \left(\frac{1}{r_{\pi1}} + g_{m1} \right) = 0 \rightarrow V_i \approx V_{in} g_{m1} r_{\pi3} \parallel r_{\pi1} \parallel \frac{1}{g_{m1}} \approx 0.5 V_{in}$$

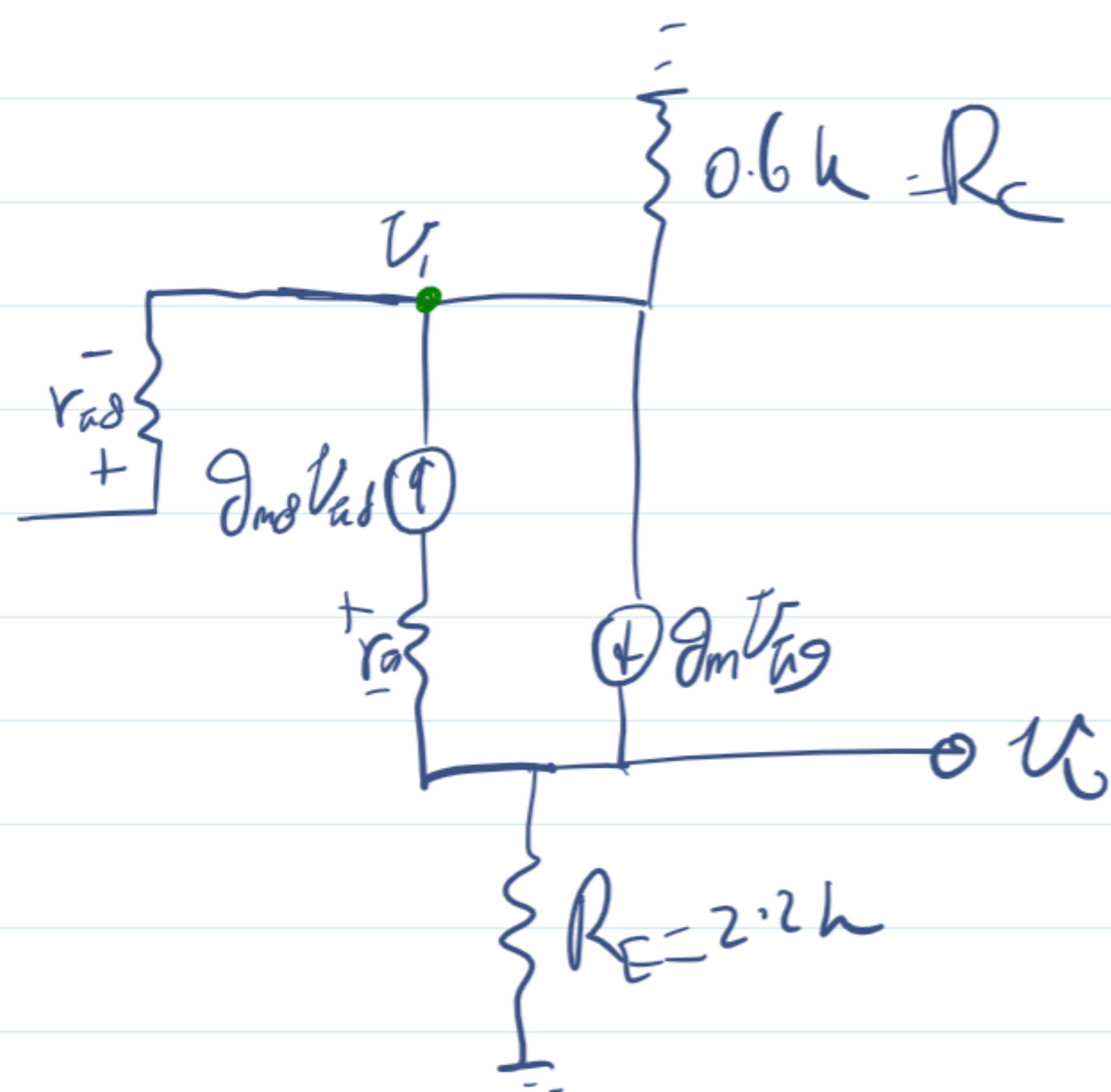
$$A_1 = \frac{V_{out}}{V_{in}} = - (g_{m1} + g_{m3}) \frac{R_c}{2} \approx g_{m3} \frac{R_c}{2} = - \frac{g_{m3}}{2} (5.6k \parallel r_{\pi6}) = -105.66$$

$$A_2 = - \frac{3k \parallel (r_{\pi7} + \beta_p (1.1k))}{r_{m6}} \approx -120$$

: A_3 جاب

$$A_3 = - \frac{10k \parallel (r_{\pi 8} + \beta(0.6k))}{r_{m7} + R_E} \approx 8.46$$

: A_4 جاب



$$V_{\pi 9} = -r_{\pi 9} g_{m8} V_{\pi 8}$$

$$V_{out} = R_E (g_{m9} V_{\pi 9} - g_{m8} V_{\pi 8}) = -g_{m8} R_E V_{\pi 8} (\beta_n + 1)$$

$V_{in} - V_{\pi 8}$

$$\widetilde{V}_i \left(\frac{1}{R_C} + \frac{1}{r_{\pi 8}} \right) - \frac{V_{in}}{r_{\pi 8}} + \underbrace{g_{m9} V_{\pi 9} - g_{m8} V_{\pi 8}}_{-g_{m8} V_{\pi 8} \beta_n} = 0$$

$$\rightarrow V_{in} \left(\frac{1}{R_C} \right) = V_{\pi 8} \left(g_{m8} \beta_n + \frac{1}{R_C} + \frac{1}{r_{\pi 8}} \right)$$

$$A_4 = \frac{V_{out}}{V_{in}} = -g_{m8} R_E \beta_n \frac{1/R_C}{g_{m8} \beta_n + 1/R_C + 1/r_{\pi 8}} \approx -3.64$$

$$A_v = 390 \times 10^3$$